

FINAL YEAR DEFENCE PRESENTATION



Design and Implementation of a School Attendance Management System

Name: TEGUE NONO MIKEL MODEIRO

Matricule: FE21A321

Supervisor: Dr. ALINE TSAGUE



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The background is a solid blue color. It features several decorative elements: a large light blue circle in the top right corner, a smaller light blue circle in the bottom left corner, and several smaller solid blue circles scattered throughout. In the upper center, there are three vertical bars of varying heights and shades of blue. The number '01' is displayed in white, bold, sans-serif font, centered within the middle bar.

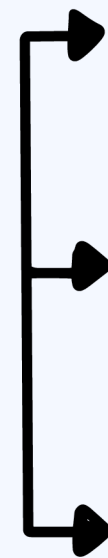
01

Background & Context

Objectives



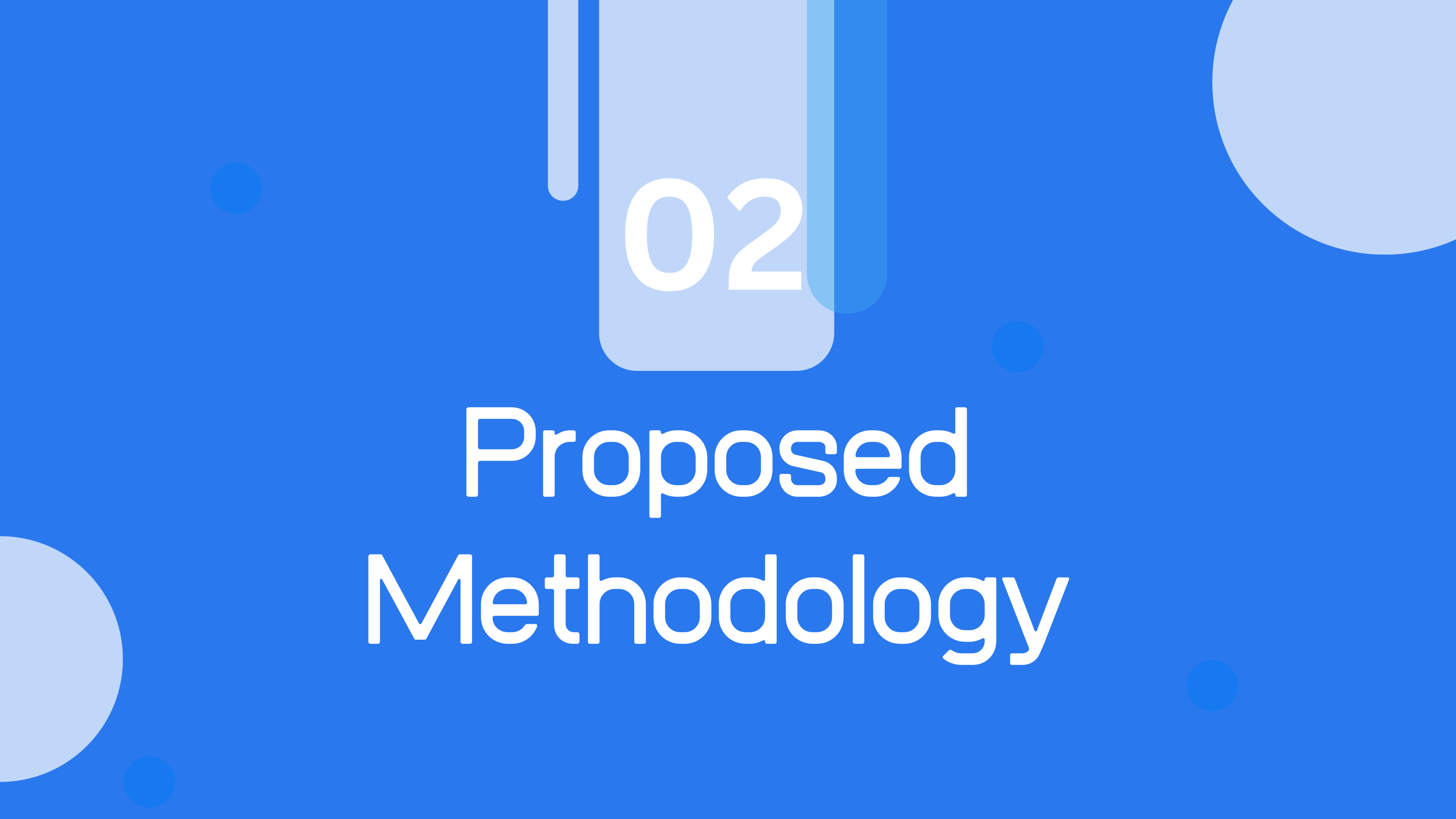
Develop an autonomous attendance management system that uses geofencing and decentralized verification to reduce fraud and improve accuracy



user-centered Ui

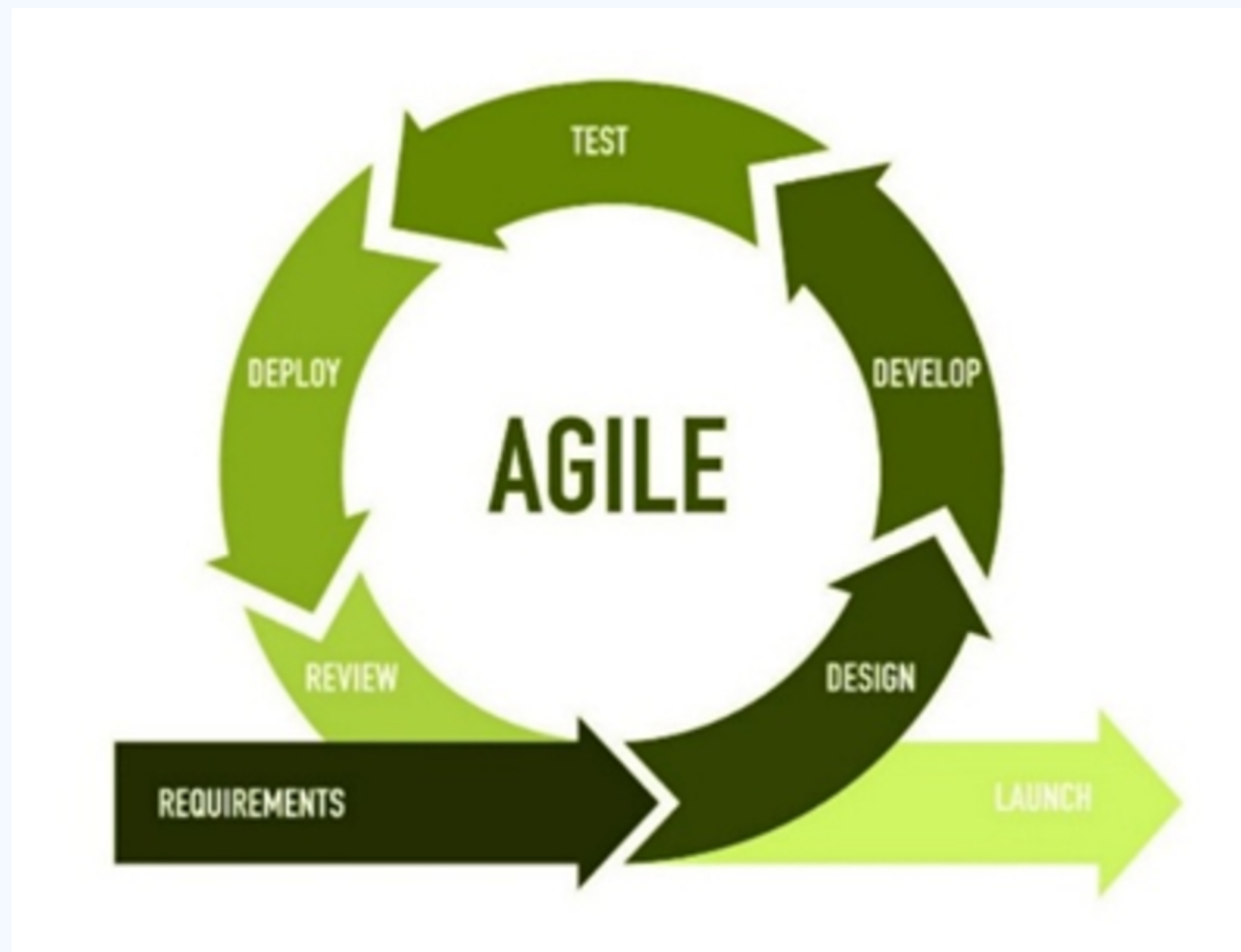
geofence verification

Decentralized record

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02

Proposed Methodology



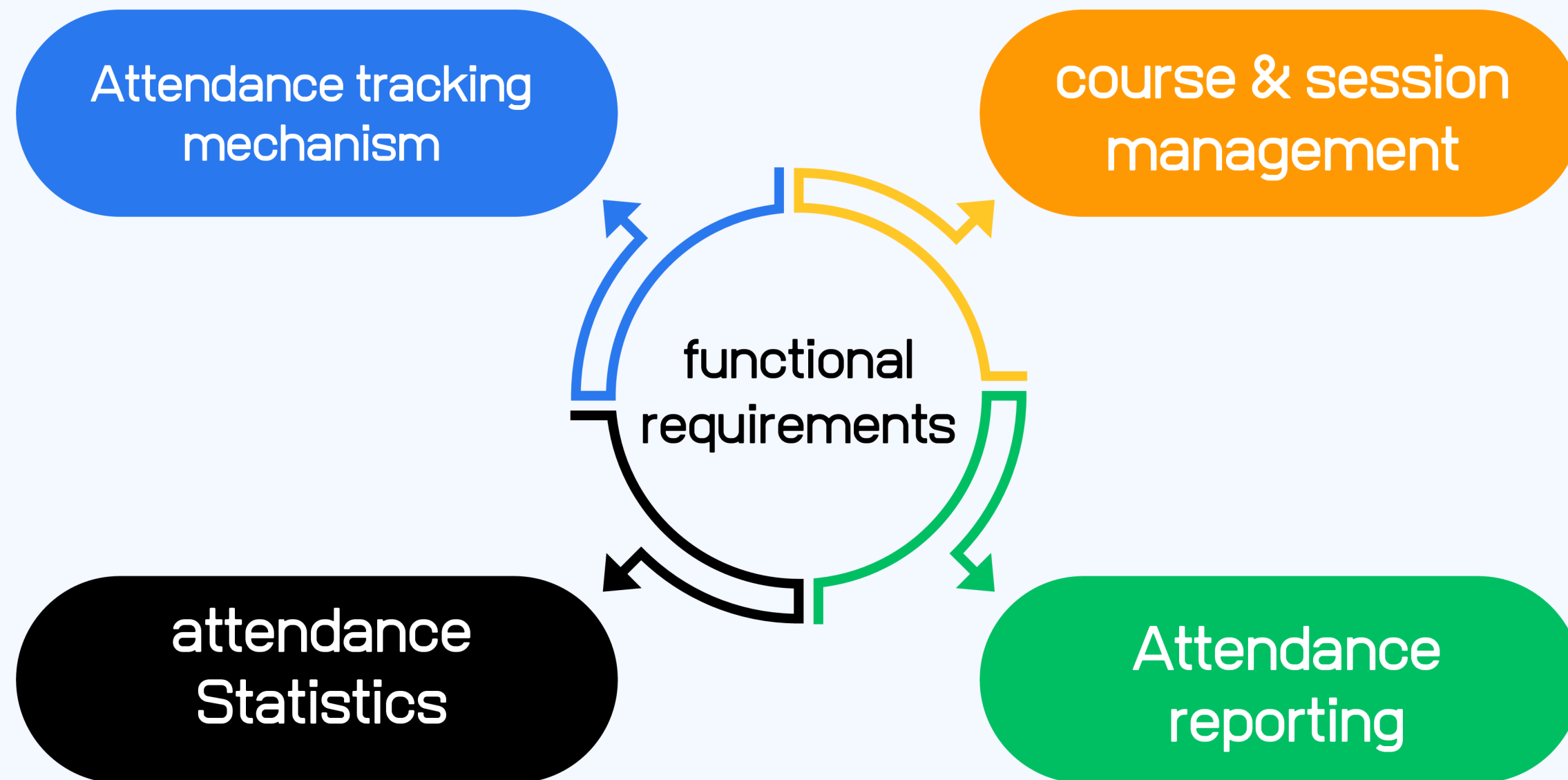
Agile

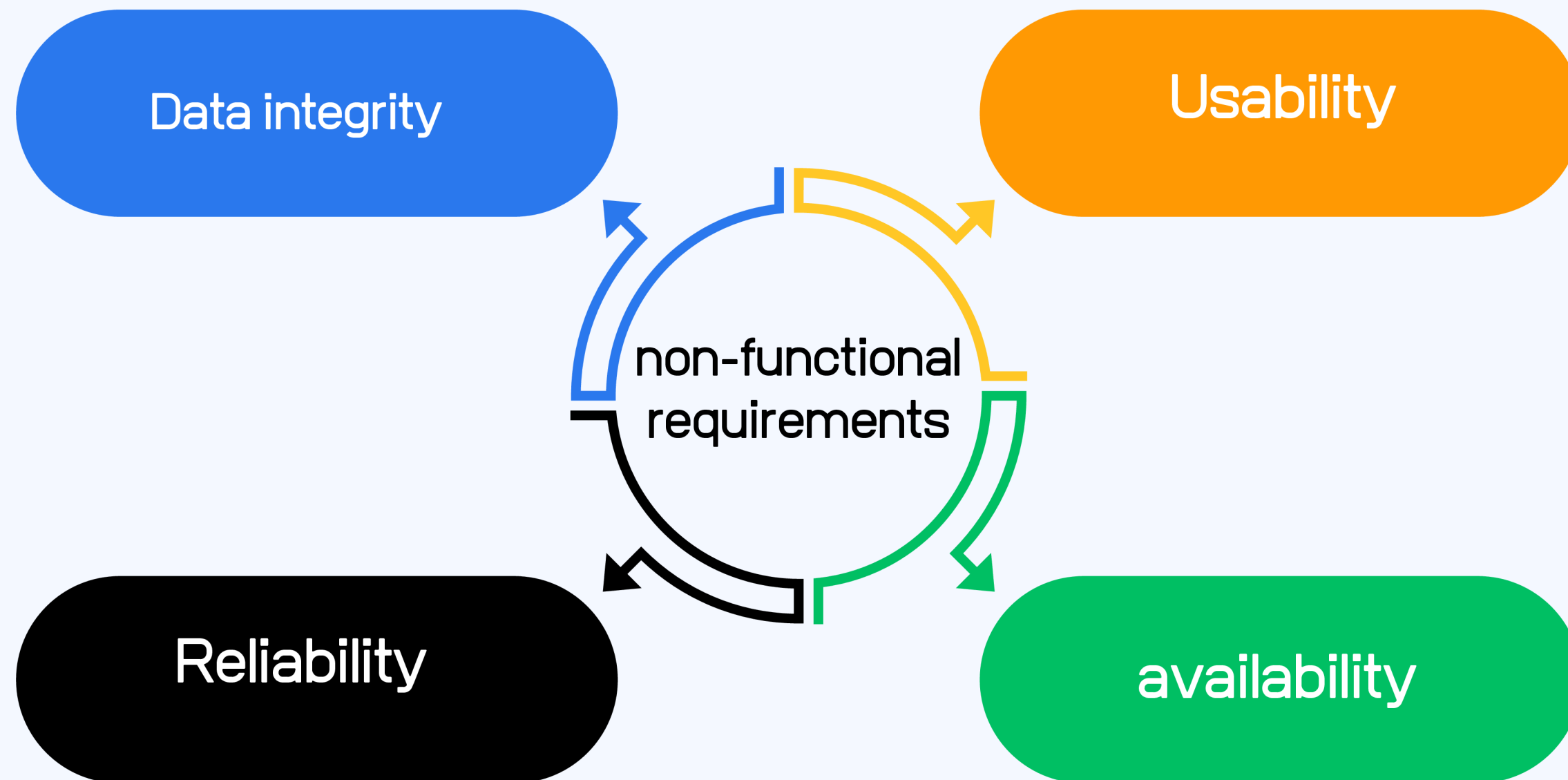
Early delivery

Flexibility

Responsiveness

The Agile philosophy promotes the rapid development and deployment of modular components.

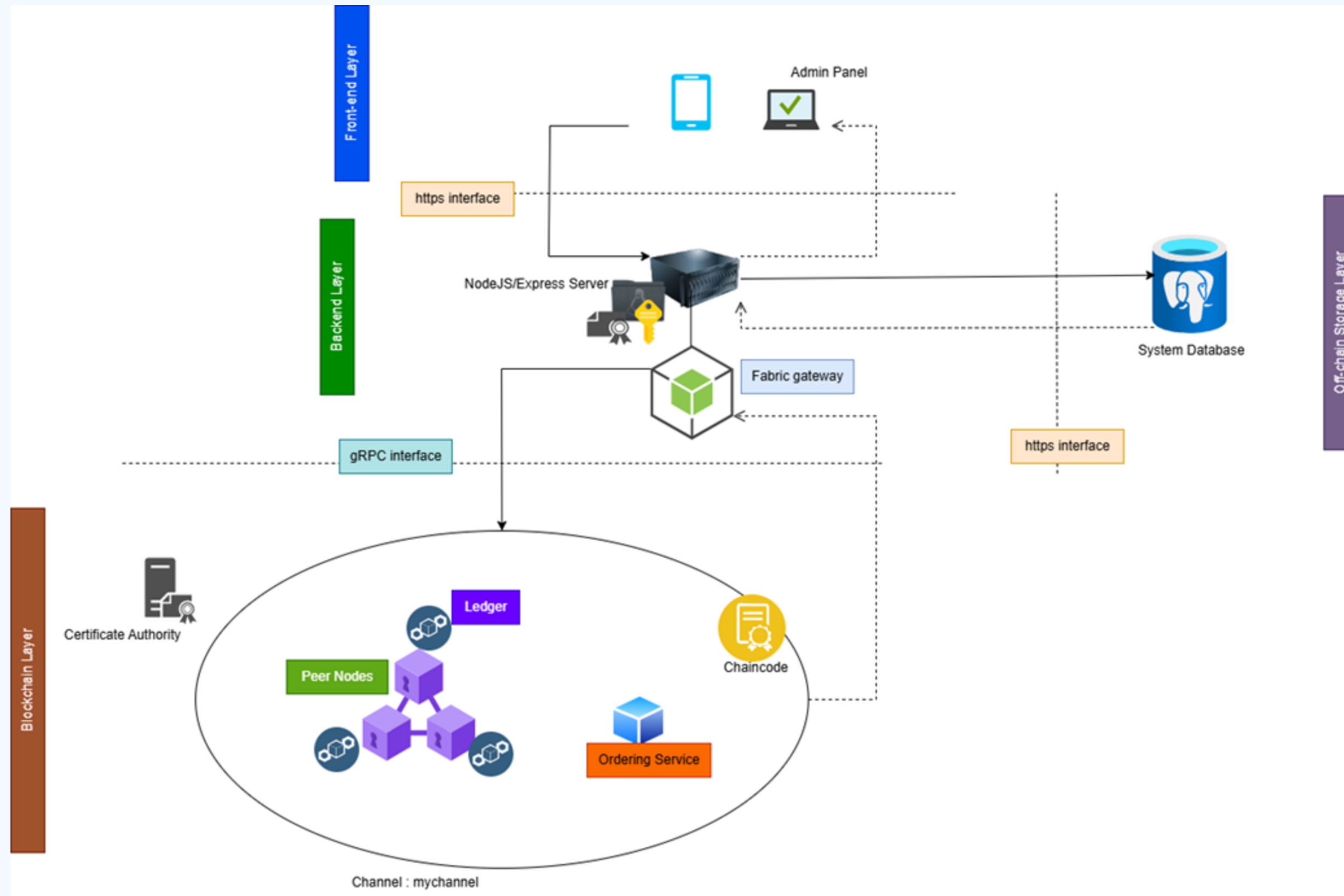




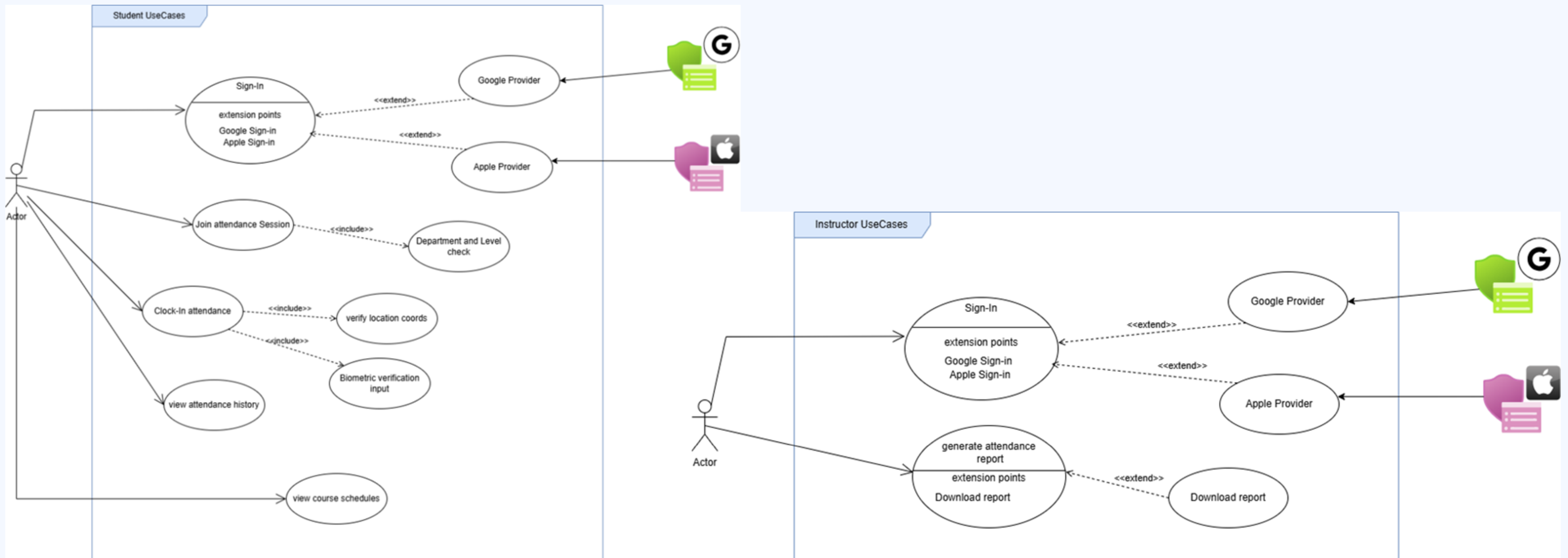
03

System Design

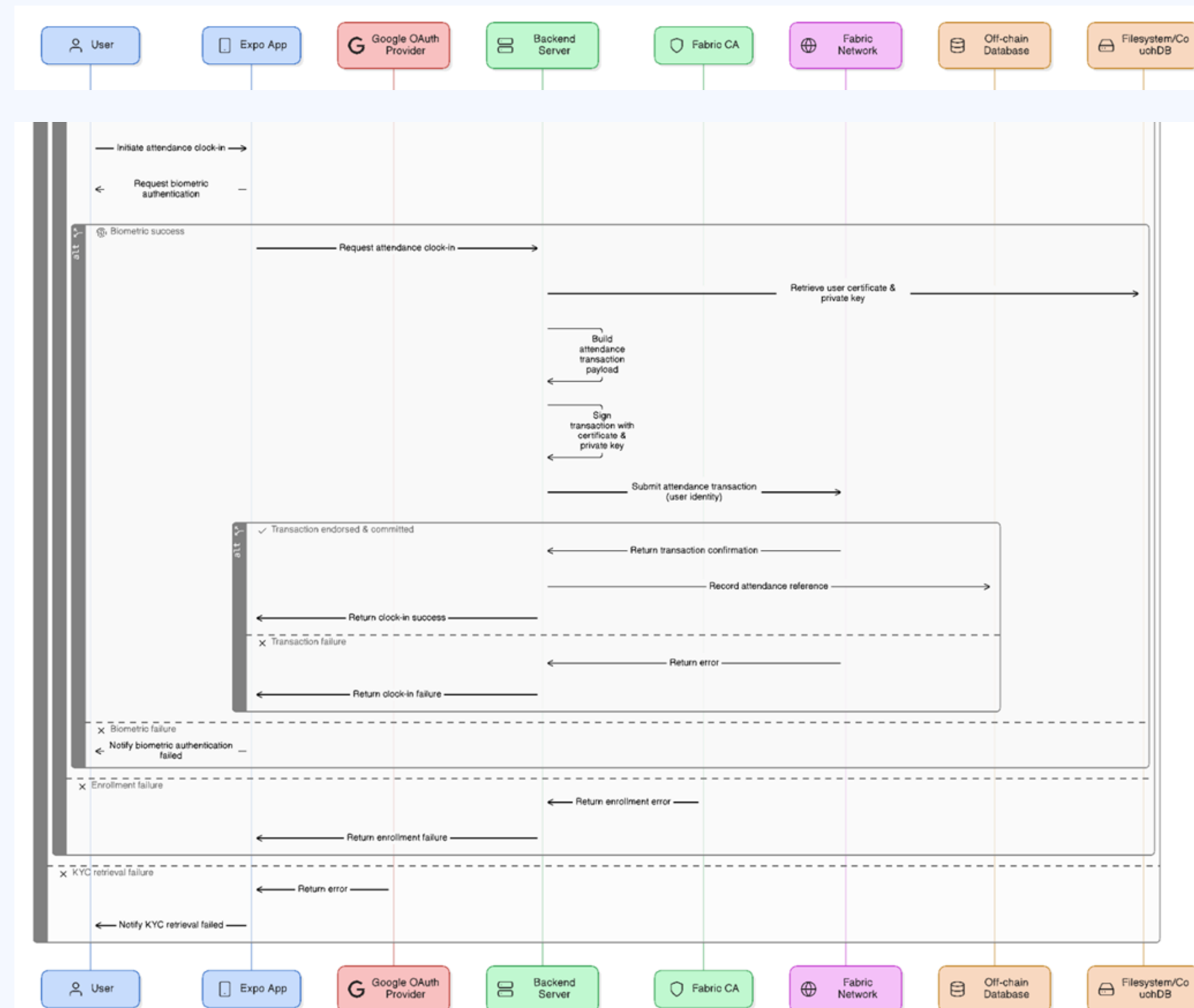
Global Architecture



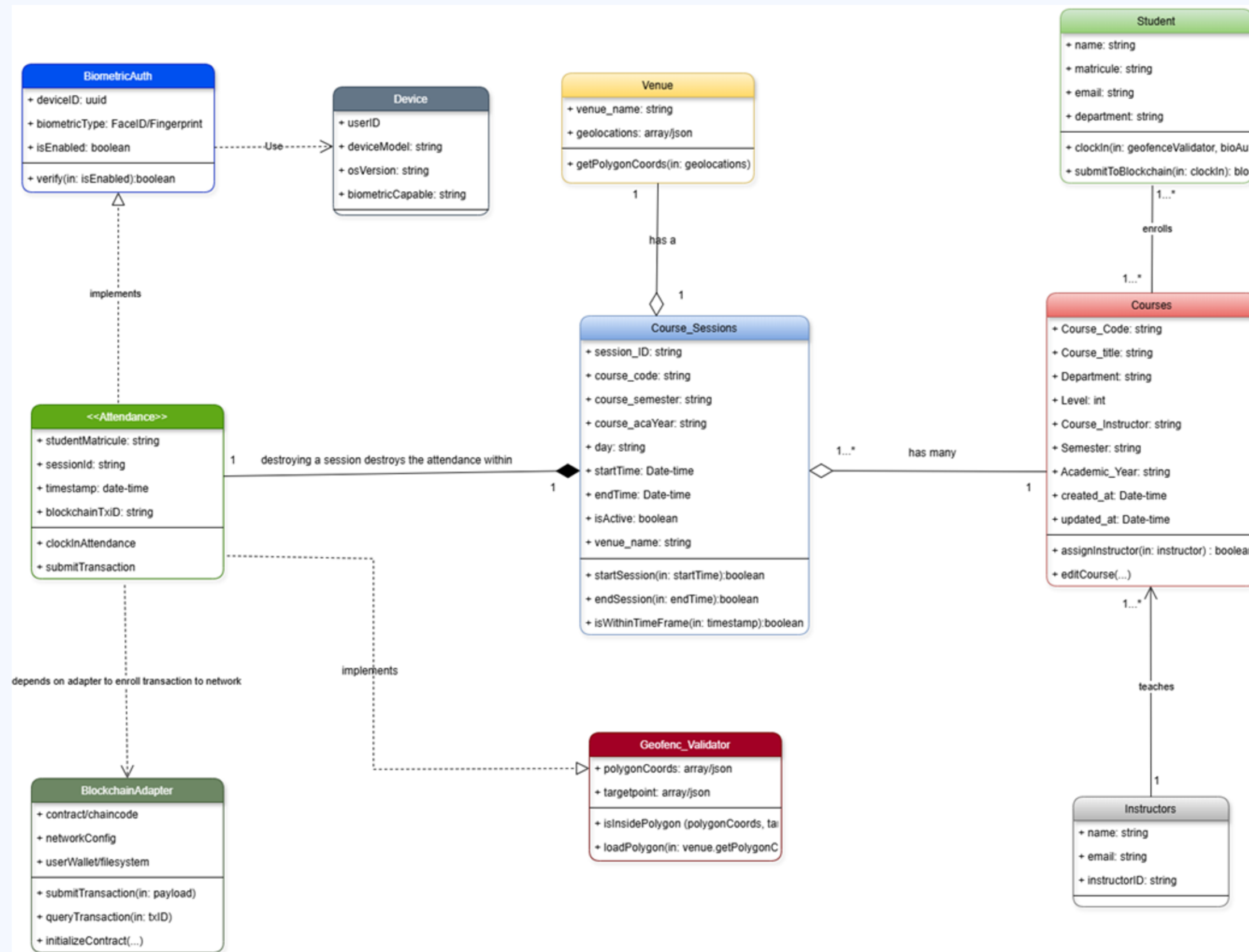
Use Case Diagrams



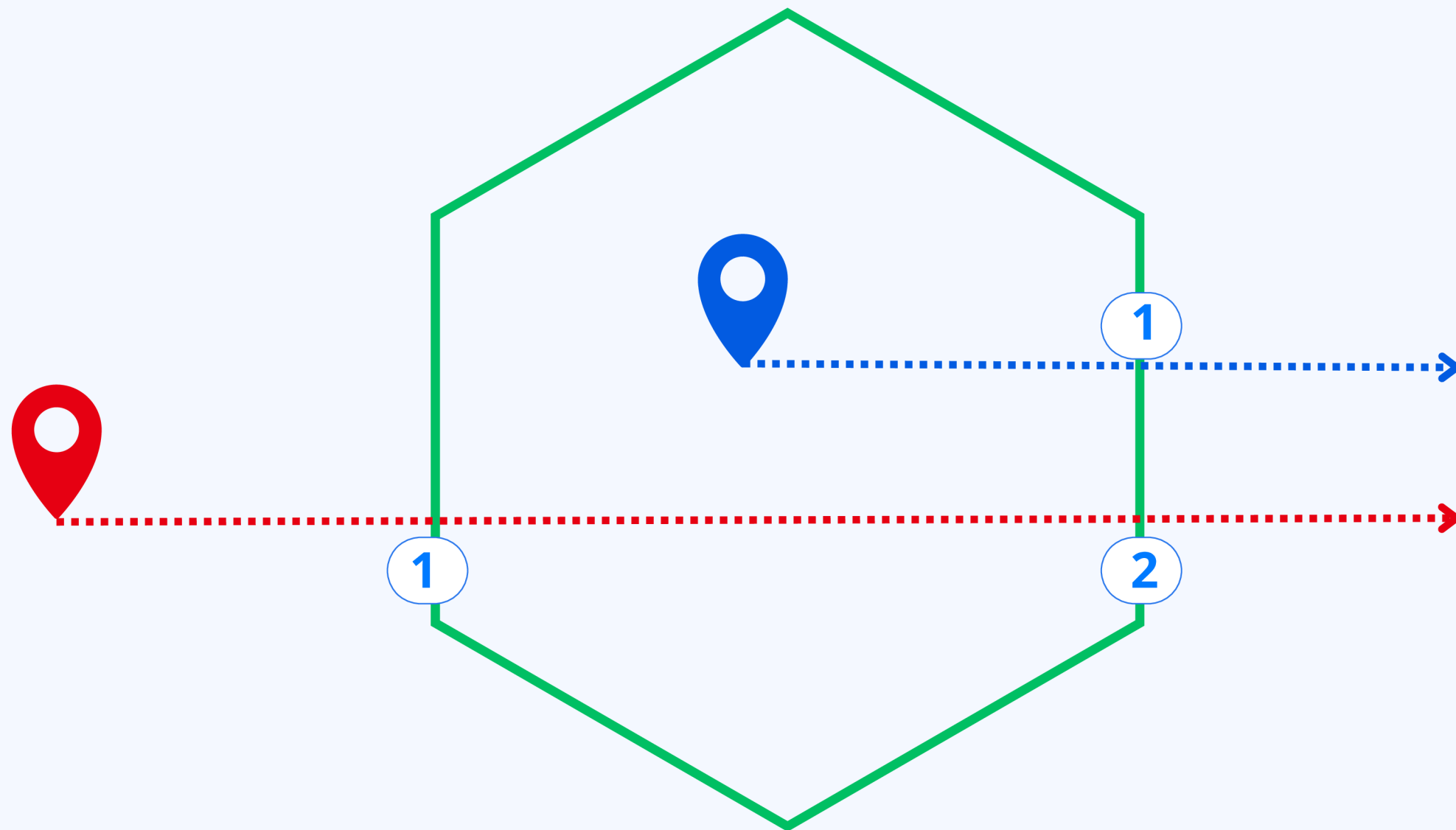
Sequence diagram



Class diagram



Geofencing Analysis & Design

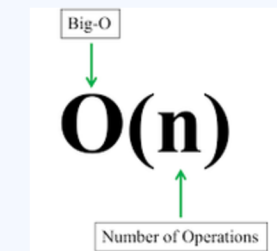


Intersection Condition

- The ray's y-coordinate y lies between $Y1$ and $Y2$ or ($Y2$ and $Y1$)
- The x-coordinate of the intersection $X_{int} = X1 + \frac{(y - Y1)(X2 - X1)}{Y2 - Y1}$ is greater than X



Time Complexity



04

Implementation and Results

Implementation

Tools | Technologies used

Tools Used



Client-Side



Backend



Blockchain Network

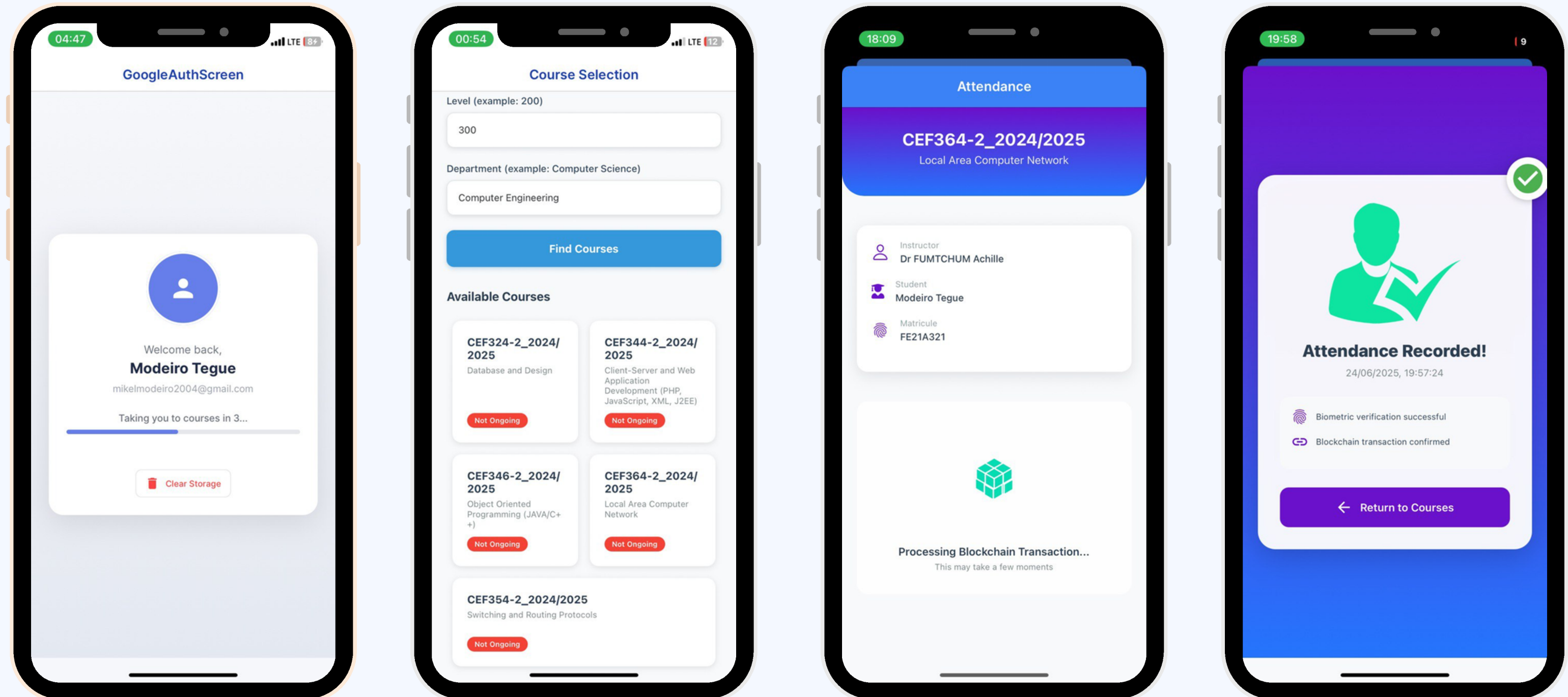


Off-chain DB



Results

Main Attendance screen

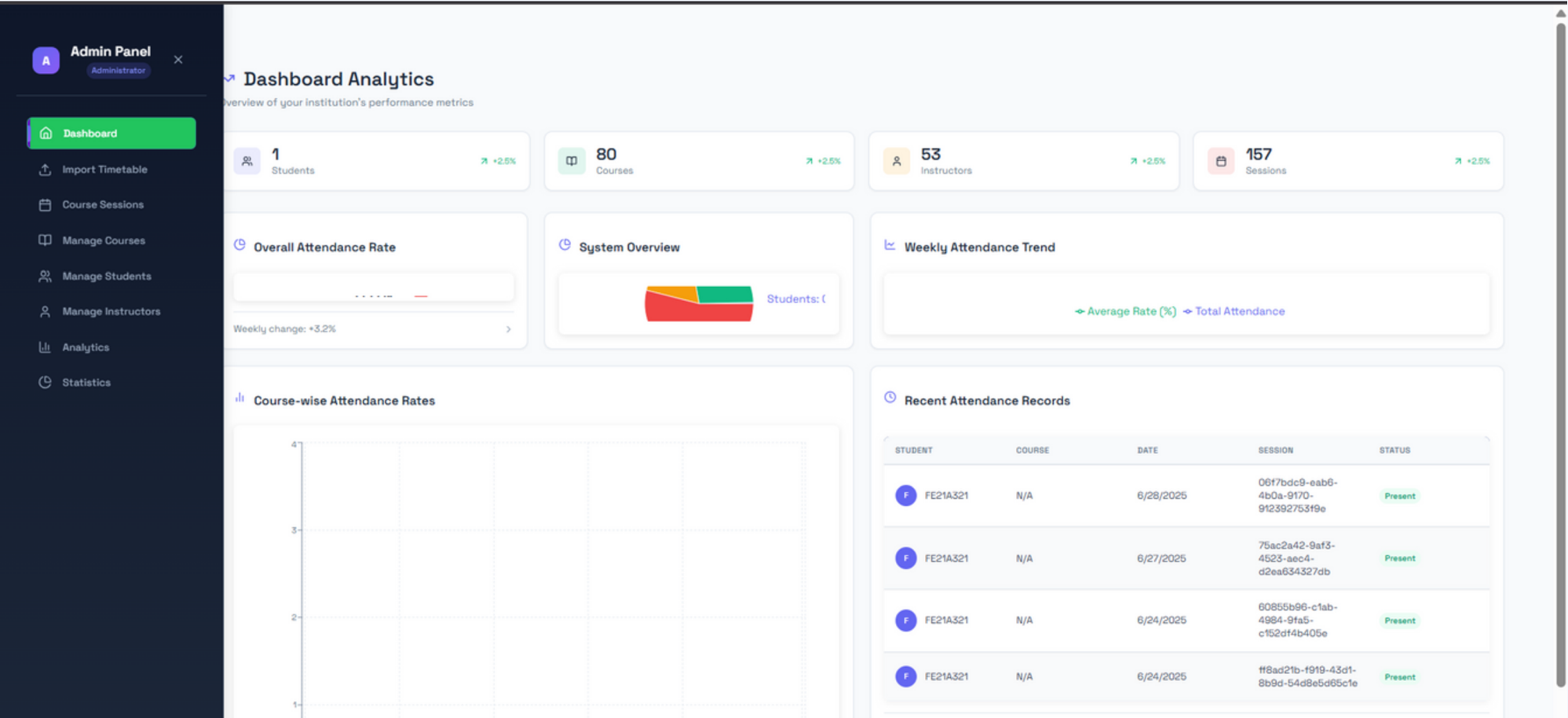


Results

Network monitoring | Admin panel

```
Chaincode initialization is not required
modeiro@MODEIRO-TEGUE:~/blockchain-network/fabric-samples/test-network$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
74e62f331e5d   dev-peer0.org1.example.com-basic_1.0-01cada8444eef79554e3ba20cab41fdeaa20a354987f7aecca6dd341d1946428  "docker-entrypoint.s..." 4 minutes ago  Up 3 minutes
dc896daa8385740bcd0 0.org1.example.com-basic_1.0-01cada8444eef79554e3ba20cab41fdeaa20a354987f7aecca6dd341d1946428  "docker-entrypoint.s..." 4 minutes ago  Up 3 minutes
1a4973a09aba     dev-peer0.org2.example.com-basic_1.0-01cada8444eef79554e3ba20cab41fdeaa20a354987f7aecca6dd341d1946428  "docker-entrypoint.s..." 4 minutes ago  Up 3 minutes
bbd6d0810d24c7f371dd 0.org2.example.com-basic_1.0-01cada8444eef79554e3ba20cab41fdeaa20a354987f7aecca6dd341d1946428  "peer node start"        8 minutes ago  Up 8 minutes  0.0.0.0:7051->7051/tcp, 0.0.0.0:9444->9444/tcp
g1.example.com   8ee7c7d1fb88   hyperledger/fabric-peer:latest "peer node start"        8 minutes ago  Up 8 minutes  0.0.0.0:9051->9051/tcp, 7051/tcp, 0.0.0.0:9054->9054/tcp
g2.example.com   9d2183590872   hyperledger/fabric-orderer:latest "orderer"              8 minutes ago  Up 8 minutes  0.0.0.0:7050->7050/tcp, 0.0.0.0:7053->7053/tcp
example.com      2cc167bea836   hyperledger/fabric-ca:latest  "sh -c 'fabric-ca-se..." 10 minutes ago  Up 9 minutes  0.0.0.0:7054->7054/tcp, 0.0.0.0:1705->1705/tcp
26d4437143bd     hyperledger/fabric-ca:latest  "sh -c 'fabric-ca-se..." 10 minutes ago  Up 9 minutes  0.0.0.0:8054->8054/tcp, 7054/tcp, 0.0.0.0:8055->8055/tcp
2a4962cf6e47     hyperledger/fabric-ca:latest  "sh -c 'fabric-ca-se..." 10 minutes ago  Up 9 minutes  0.0.0.0:9054->9054/tcp, 7054/tcp, 0.0.0.0:9055->9055/tcp
```

```
modeiro@MODEIRO-TEGUE:~/blockchain-network/fabric-samples/test-network$ ./network.sh deployCC -ccn basic -l javascript
Using docker and docker-compose
deploying chaincode on channel 'mychannel'
executing with the following
- CHANNEL_NAME: mychannel
- CC_NAME: basic
- CC_SRC_PATH: ../attendance/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
- CC_SEQUENCE: auto
- CC_END_POLICY: NA
- CC_COLL_CONFIG: NA
- CC_INIT_FCN: NA
- DELAY: 3
- MAX_RETRY: 5
- VERBOSE: false
executing with the following
- CC_NAME: basic
- CC_SRC_PATH: ../attendance/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
+ '[' false = true ']'
+ peer lifecycle chaincode package basic.tar.gz --path ../attendance/chaincode-javascript --lang node --label basic
+ res=0
Chaincode is packaged
Installing chaincode on peer0.org1...
Using organization 1
+ peer lifecycle chaincode queryinstalled --output json
+ jq -r 'try (.installed_chaincodes[].package_id)'
+ grep '^basic_1.0:17b6a9c96729d3a400ed154a2fb969520b9968cc78f954217120699ee84db57d$'
+ test 1 -ne 0
```



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Conclusion

Difficulties Encountered

1

Steep Learning
curve in
understanding
Web3 fundamentals

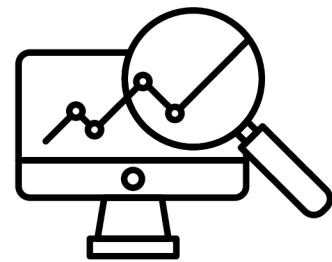
2

Hyperledger fabric
Environment setup

3

Chain-code
discovery and user
identity
management

Further Works



Real Time based
Event monitoring



Instructor
portal

Demo

